BIOCHEMISTRY CUME
FEBRUARY 27, 2004

The oxidation of succinate produced by the citric acid cycle inside the mitochondrial membrane is carried out by three membrane complexes, complex II (succinate – CoQ oxidoreductase), complex III (CoQ – cytochrome c oxidoreductase) and complex IV (cytochrome c oxidase). These three mitochondrial complexes carry out the transfer of two electrons from succinate, with a redox potential of +0.03 V, to oxygen, with a redox potential of +0.82 V. This electron transfer process is coupled to the translocation of 4 protons from the inside to the outside of the mitochondrial membrane by complexes II and III, and an additional 2 protons translocated by complex IV. Complex II contains the flavin prosthetic group, FAD, and a number of iron-sulfur centers. The electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.10$ V, and the pH inside the mitochondrial membrane was $pH_{in} = 8.5$ and $pH_{out} = 6.5$.

1. (30) Draw a diagram showing how complexes II, III, and IV are arranged in the mitochondrial membrane, and how they couple the energy of electron transfer from succinate to O_2 to pumping protons across the membrane. Show the redox centers of the complexes in your diagram. Describe what role (if any) complex II plays in proton translocation across the membrane.

2. (25) Calculate the free energy $\Delta G^{o'}$ for electron transfer from 1 mole of succinate to 0.5 mole O_2.

3. (30) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane, and calculate ΔG for proton pumping that is coupled to electron transfer from succinate to O_2.

4. (20) Calculate the efficiency with which complexes II, III, and IV couple the exergonic free energy of electron transfer to the endergonic free energy of proton pumping.

Possibly useful formulas:

$\Delta G^{o'} = -RT\ln K_{eq}$; $\Delta G^{o} = -nF\Delta E_o$

$\Delta G = \Delta G^{o} + RT\ln\{\text{products/reactants}\} + nF\Delta V$

F = Faraday's constant = 23 kcal/volt; $R = 1.987 \text{cal mol}^{-1} \text{deg}^{-1}$